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Robotics in Healthcare

Applications in Surgery, Rehabilitation, Assistive Care, and Hospital Services

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1. Introduction

Robotics combines mechanical systems, sensing, control, computation, and increasingly artificial intelligence to perform physical tasks with repeatability and precision. Healthcare is an important application area because clinical work contains tasks that require accurate movement, repetitive training, controlled manipulation, or continuous monitoring. Modern medical robotics includes surgical systems, rehabilitation robots, assistive and collaborative robots, hospital service robots, and robotic devices that interact with sensors and digital health platforms. A 2024 survey describes healthcare robotics as a rapidly growing field with potential benefits for patient care, clinical productivity, and safety for both patients and healthcare workers [1].

2. Surgical Robotics

Robot-assisted surgery is one of the most visible healthcare robotics applications. A surgical robotic system normally combines a surgeon interface, robotic manipulators, cameras or other sensors, control software, and specialized instruments. In a teleoperated arrangement, the surgeon remains responsible for the procedure while the robotic mechanism translates the surgeon's commands into controlled instrument motion. This can support precise manipulation in minimally invasive procedures and may provide enhanced visualization and instrument articulation.

Recent reviews show that commercially available robotic surgical systems are being investigated or used across several specialties, including soft-tissue surgery, orthopedics, neurosurgery, and microsurgery [2]. The important engineering challenge is not simply making a robot move; the system must provide accurate control, safe limits, reliable sensing, appropriate human-machine interfaces, and clinically validated performance. Training is also important because clinicians need to understand both the surgical procedure and the specific control interface of robotic platforms [3].

3. Rehabilitation Robotics

Rehabilitation robotics uses controlled robotic assistance to help patients perform therapeutic movements. Systems may target the upper limb, lower limb, gait, or specific functional movements. A major advantage is that a robotic device can deliver repeated and measurable exercises while adjusting assistance or resistance.

A 2024 systematic review reported that robotic technologies such as exoskeletons and assistive training systems have been studied for patients with physical disabilities and can support functional recovery, motor function, strength, coordination, and dexterity [4]. Another 2024 umbrella review synthesized 62 systematic reviews covering 341 randomized controlled trials and 14,522 participants, with stroke being the most frequently represented condition [5]. These findings illustrate why rehabilitation robots are attractive: they can provide structured repetition, quantitative measurements, and controlled assistance while allowing therapists to supervise treatment.

4. Assistive and Collaborative Robots

Assistive robots are designed to support people with reduced mobility or limited ability to perform activities of daily living. Examples include robotic feeding aids, mobility assistance, exoskeletons, and systems that help users interact with their environment. Collaborative robots, or cobots, can also support healthcare workers rather than replacing them. They may assist with routine logistical activities, monitoring, medication delivery.

A 2024 systematic review of collaborative robots for nurses found applications including medication delivery, vital monitoring, and social interaction, while also noting that the evidence base for real-world deployment remains limited [6]. This highlights a central principle of healthcare robotics: successful systems should be designed around human needs and workflows. Human oversight, usability, safety, and acceptance are as important as mechanical performance.

5. Hospital Service and Disinfection Robots

Hospitals also use mobile robotic systems for non-surgical tasks. Potential applications include transportation of supplies, delivery of medicines or meals, navigation assistance, patient interaction, and room disinfection. These robots can reduce exposure to repetitive or hazardous tasks and help organize hospital workflows. Reviews of medical service robots describe applications ranging from patient support and telepresence to sanitization [7].

For such robots, autonomous navigation is especially important. The robot must detect obstacles, understand its environment, plan a safe route, and respond when people or equipment unexpectedly enter its path. This makes sensors, mapping, localization, motion planning, and human-robot interaction central parts of the system design.

6. Soft Robotics in Healthcare

Soft robotics uses compliant materials and flexible structures instead of relying only on rigid links and joints. This approach is attractive for healthcare because many interactions involve delicate human tissue. Soft robotic devices have been explored for rehabilitation, wearable assistance, surgical applications, and other biomedical uses [8]. Their flexibility can make human interaction more comfortable and can allow devices to conform to body-shape.

However, soft robotics introduces engineering challenges including modeling, control, durability, sensing, biocompatibility, and long-term reliability. These issues must be considered before a prototype can be translated into a clinically dependable medical device.

7. Benefits of Robotics in Healthcare

The major potential benefits of healthcare robotics include precision, repeatability, controlled motion, quantitative measurement, assistance with repetitive tasks, and improved access to certain services. Rehabilitation robots can deliver repeatable exercises, while surgical systems can provide highly controlled instrument motion. Mobile robots can support logistics and service tasks. In addition, robotic platforms can collect sensor data that may help clinicians evaluate performance or monitor a process.

From an engineering perspective, robotics also creates opportunities to integrate sensors, embedded systems, communication networks, computer vision, artificial intelligence, and data analytics into a single healthcare platform.

8. Challenges and Limitations

Healthcare robotics is safety-critical. A failure that would be inconvenient in an ordinary industrial application can become harmful in a clinical environment. Important challenges include high development and maintenance costs, regulatory requirements, cybersecurity, reliability, human factors, training, and the need for clinical evidence. AI-enabled medical robots add further concerns involving transparency, robustness, trust, and accountability.

The literature also emphasizes that many promising technologies remain at research or proof-of-concept stages. Practical clinical adoption requires controlled evaluation, standardized performance criteria, integration with existing workflows, and clear responsibility between humans and machines [7].

9. Future Scope

The future of healthcare robotics is likely to involve greater integration between robotics, artificial intelligence, computer vision, wearable sensors, Internet of Things technologies, and digital twins. AI can support perception, planning, adaptive rehabilitation, and human-robot interaction, while IoT can connect robots to sensor or edge platforms.

Importantly, future systems are more likely to emphasize human-robot collaboration than complete replacement of healthcare professionals. Recent work on clinical human-robot collaboration argues for bounded autonomy in which clinicians retain responsibility while robots support sensing, planning, and task execution within defined safety limits [9].

10. Conclusion

Robotics is becoming an important engineering technology in healthcare, with applications ranging from surgical assistance and rehabilitation to hospital logistics and assistive care. The strongest value of a healthcare robot is not simply autonomous movement; it is the ability to perform a useful clinical or support task accurately, repeatably, measurably, and safely. Biomedical engineers have an important role in this field because they can connect mechanical design and control with physiology, medical workflows, sensors, and patient requirements. Continued progress will depend on multidisciplinary research, clinical validation, safety engineering, and responsible human-centered design.

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